

Seydi Cheikh Wade

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EDUCATION

Carleton University

Ottawa, ON

B.Eng. Software Engineering (Co-op) – GPA: 3.6/4.0

Graduating April 2028

- Coursework: Database Systems, Algorithms & Data Structures, Operating Systems, OO Software Development, Real-Time Concurrent Systems, Requirements Engineering, Computer Architecture

WORK EXPERIENCE

Open-Source Statistical Programmer (Co-op)

May 2026 – Aug. 2026

Statistics Canada

Ottawa, ON

- Contributing to the **SAS-to-Python migration** of legacy statistical programs, applying **Python** and **pandas** to modernize data processing workflows for a federal government statistical agency
- Developing and maintaining **ETL pipelines** for large-scale statistical data processing, ensuring data integrity and consistency across transformation stages in preparation for downstream analysis

Junior Data Scientist (Co-op)

Jun. 2025 – Sep. 2025

Jasmine Conseil – Federal Government Client

Laval, QC

- Built an **ML prediction system** for the Senegalese Customs Administration to automate control circuit assignment across **10,000+ import declarations**, directly supporting **40% of national budget operations**
- Benchmarked 5+ classifiers; selected **MLPClassifier+PCA** achieving the highest **ROC-AUC (0.99)** and **F1-score (0.86)** among all evaluated models on imbalanced data; serialized with Joblib for production serving
- Deployed the trained model as a **FastAPI** REST API on a **DigitalOcean Linux VM** with Pydantic validation and Swagger docs; configured **firewalld** rules and routed traffic through a **MuleSoft Anypoint proxy** for rate limiting

PROJECTS

ChizuCode – Hackathon Winner ↗ | Next.js, FastAPI, PostgreSQL, pgvector, Gemini API, Voyage AI

- Won MLH Best Use of Gemini API** at ConHacks 2026 with a team of 4; built an **end-to-end RAG pipeline** that ingests GitHub repositories, generates dual embeddings (**Gemini** for summaries, **Voyage AI** for code), stores vectors in **pgvector**, and clusters files into semantic domains using **scikit-learn hierarchical clustering**
- Implemented **scoped retrieval**: embeds user questions, performs **pgvector similarity search** over chunked code and summaries, applies relevance filtering, and synthesizes source-backed answers with **Gemini** grounded in retrieved chunks

Soki – Hackathon Winner ↗ | Next.js, FastAPI, PostgreSQL, Gemini API, FSRS

- Won 1st place** (AI Automation) and **runner-up** (Best Language Cognition) at Carleton MindHack Hackathon, building an **AI-powered study platform** in a **36-hour sprint** with a team of 3
- Designed a **content generation pipeline** that parses uploaded PDFs, sends structured prompts to **Gemini API**, and returns flashcards and quizzes tailored to source material; integrated an **FSRS-based spaced repetition scheduler** that adapts review intervals to individual learning patterns

Firefighting Drone Swarm Simulator ↗ | Java, UDP, Concurrency, State Machines

- Built the **scheduler** for a distributed drone coordination system running as **3 separate Java processes** communicating over **UDP (DatagramSocket)**, dispatching multiple drones to fire zones in real time
- Implemented **fault detection and recovery**: scheduler detects drones stuck mid-flight or jammed nozzles via timing events, reroutes missions to available drones, and marks faulted units offline

EXTRACURRICULAR & VOLUNTEERING

Software Engineer (Volunteer)

May 2026 – Aug. 2026

Develop for Good – Crisis Center, Inc.

- Built and redesigned a website for a crisis support nonprofit serving clients, donors, and volunteers; contributed to UX research, information architecture, and low/no-code implementation as part of a structured volunteer engineering team

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, C++, JavaScript, TypeScript

ML & Data Science: scikit-learn, pandas, NumPy, supervised learning (MLP, XGBoost, SVM, Decision Trees), PCA, feature engineering, F1-score/ROC-AUC optimization, imbalanced data handling

Frameworks & APIs: FastAPI, REST APIs, Pydantic, Joblib, Next.js, NextAuth, JDBC, Jupyter

Databases: PostgreSQL, pgvector

Developer Tools: Git, GitHub, Bitbucket, Docker, Linux CLI, Jira, Confluence

Cloud & Infrastructure: DigitalOcean, MuleSoft Anypoint